Asaad Hanna

Quantum Simulation | Electronic Structure | Scientific Computing

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Research Profile

Physics MSc student researching quantum-algorithm accuracy and resource costs, with hands-on experience computing many-body Green's functions for Hubbard models. Background in fermionic mappings, variational state preparation, symmetry-based qubit reduction, and numerical validation. Interested in quantum algorithms for electronic structure and strongly correlated systems.

Education

MSc in Physics
Concordia University, Montreal — Sep. 2025 – Aug. 2027 (expected)
Research in quantum algorithms; supervisor: Dr. Abhijeet Alase.

BSc, Honours in Physics, with Distinction
Concordia University, Montreal — 2022 – 2025
Minor in Computer Science.

Research Experience

Quantum Software Developer Intern
AlgoLabs, Université de Sherbrooke | Dr. Simon Verret — May – Aug. 2026

- Extended a Liouvillian-recursion Green's-function workflow to three- and five-site Hubbard models with degenerate ground states.
- Implemented fermionic-swap circuits connecting spin-related ground states; combined pure-state contributions to evaluate ensemble observables without directly preparing a mixed state.
- Compared Green's functions and Galitskii–Migdal energy estimates with exact-diagonalization benchmarks, using Qiskit simulators and IBM quantum hardware.
- Investigated symmetry-based qubit tapering and implemented a symplectic-method demonstration to explore qubit reduction.

Research code: Liouville-Lanczos, asaad branch.

MSc Research — Quantum Algorithms for Operator Functions
Concordia University | Dr. Abhijeet Alase — 2025 – present

- Investigating accuracy and resource costs of algorithms that implement functions of unitary operators, a primitive used across quantum applications.
- Comparing Laurent approximation and circuit constructions through mathematical analysis and computational benchmarks, distinguishing approximation error, query complexity, and implementation overhead.
- Formulating bounds and optimality questions for different function classes and applications; these are ongoing research directions.

Honours Thesis — Statistical Mechanics and Machine Learning
Concordia University | Dr. Saurabh Maiti — 2024 – 2025

- Built numerical demonstrations of equilibration and entropy in statistical ensembles; explored connections between statistical mechanics and Boltzmann machines.
- Developed simulations and visualizations to explain equilibrium distributions and learning dynamics.

Additional Research Experience

Optical Microresonators
Concordia University | Dr. Pablo Bianucci 2023 – 2024

- Performed fabrication, experiments, measurements, and data analysis for a systematic study of flame-fabricated surface nanoscale axial photonic resonators; first author of the resulting 2024 conference publication.

Research courses: Twisted bilayer graphene fabrication with Dr. Alexandre Champagne (2025); symmetries of Hermitian matrices with Dr. Saurabh Maiti (2024), Concordia University.

Publications

Published conference paper
A. Hanna, E. C. V. Eadie, S. Deep, and P. Bianucci. Systematic study of Flame-Fabricated Surface Nanoscale Axial Photonics resonators. *Advanced Photonics Congress 2024*, paper SoTh3D.3 (Optica, 2024). doi:10.1364/SOF.2024.SoTh3D.3.

Journal manuscript under review
S. Padvaiskas, E. Eadie, S. Sherr, **A. Hanna**, S. Deep, and P. Bianucci. Systematic Study of Flame-heating Fabrication of Surface Nanoscale Axial Photonic Resonators. *Advanced Optical Technologies*, submitted September 2026; under review.

Teaching Experience

Teaching Assistant — Physics
Concordia University Jan. 2025 – present

- Held successive appointments each semester: Electricity and Magnetism grading for approximately 150 students; quiz writing for Energy and Environment; advanced laboratory instruction and report assessment.
- Currently lead Electricity and Magnetism tutorials, guiding students through calculations and physical reasoning.

Awards & Research Funding

Hydro-Québec Graduate Award (2025–2026); **NSERC** Undergraduate Summer Research Award (2025); **Concordia** undergraduate research awards (2023, 2024).

Fields Institute visitor grant awarded for the 2026 thematic program on quantum algorithms for differential equations; not taken up.

Technical Skills & Training

Scientific computing: Python, NumPy, Qiskit, Qiskit Nature, PennyLane; exact diagonalization, fermion-to-qubit mappings, and numerical benchmarking.

Additional tools: Mathematica, MATLAB, Git, Linux, Bash, LaTeX, and Manim.

Training: AlgoLabs Summer Course on Quantum Algorithms, Université de Sherbrooke (2025).

Languages: English, French, and Arabic.

Leadership & Scientific Communication

Founder and Lead, Quantum Computing Club (2025–2026): delivered quantum-algorithm workshops, prepared undergraduates for hackathons, and organized researcher interviews and outreach.

Presented *The Equilibrating Behaviour of Statistical Systems* at INTRIQ and the Concordia Undergraduate Research Showcase (2025); additional poster presentation at AQC UdeS.